



Design Document (DD)

Progetto di Ingegneria del Software 2, 2019/2020

Version 1.0

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Design Document (DD)	0
1. Introduction	2
1.1 Purpose	2
1.2 Scope	2
1.3 Definitions	2
1.3.1 Definitions	2
1.3.2 Acronyms	3
1.3.3 Abbreviations	3
1.4 Revision History	3
1.5 References	4
1.6 Document Structure	4
2. Architectural design	5
2.1 Overview	5
2.2 Component view	6
2.3 Deployment view	8
2.4 Runtime view	8
2.5 Component interfaces	9
2.6 Selected architectural styles and patterns	9
4 Tier Client-Server	9
Model View Adapter	9
Thin Client	10
Relational DB	10
Loose Coupling	10
Stateless components	10
2.7 Other design decisions	10
Google APIs	10
Municipality API	11
3. User Interface Design	12
4. Requirements traceability	13
5. Implementation, integration and test plan	16
6. Effort Spent	17

1. Introduction

1.1 Purpose

This document aims to give more technical details about the design, the implementation and the testing of the application. It is complementary to the *Requirements Analysis Specification Document*. It aims to give an unified vision of the software so that every team member that will work on it can be coordinated.

The document mainly presents:

- A high level architecture of the software
- The main components and their interfaces
- The runtime behaviour
- The design patterns
- How the requirements map to the design elements
- An implementation, integration and tests plan

1.2 Scope

SafeStreets is a crowd-sourced application that lets users notify authorities when traffic violations occur. It offers a basic service and some advanced functionalities based on the type of user.

The SafeStreets service will be offered through three different channels: a mobile application, for iOS and Android, a web application and an API. The mobile and web application will be usable by anyone upon registration, be they a user or an authority. The API will be used instead for communicating with Municipalities, a particular case of Authority.

In general, the aim of SafeStreets can be divided in two main parts: the first one is to let users notify authorities about the traffic violations they encounter, the

second is to let authorities and municipalities know better information about traffic violations encountered by citizens.

The service offered to Citizens will let them communicate with the authorities and report the parking violations they notice in streets. They will be able to send pics, time, date and position to authorities. Examples of violations are vehicles parked in the middle of bike lanes or in places reserved for people with disabilities, double parking, and so on.

On the other hand, authorities and municipalities will be able to gain a better sight on the violations that happen on the territory, through the data mining functionality and the mutual sharing of information with SafeStreets.

1.3 Definitions

1.3.1 Definitions

- **Applicant:** we define applicant as the entity who makes the requests to extract the data about parking violations
- **Authority:** it is the service's consumer which has to grant the safety of an area. For more details see section [2.3 User characteristics](#) of the RASD document.
- **Municipality:** it is a particular case of Authority. It can access a broader set of functionalities than a normal authority.
- **SafeStreets:** it is the name of the all software described by this paper. It includes different functions.
- **User:** it is the common single consumer of the service. For more details see section [2.3 User characteristics](#) of the RASD document.

1.3.2 Acronyms

- **API:** Application Programming Interface
- **DB:** Database

- **DBMS:** Database Management System
- **GPS:** Global Positioning System
- **GDPR:** General Data Protection Regulation
- **HW:** Hardware
- **RASD:** Requirement Analysis Specification Document
- **UML:** Unified modelling language
- **E-R:** Entity-Relationship

1.3.3 Abbreviations

- **[Gxx]:** goal number xx
- **[Rxx]:** requirement number xx
- **[Dxx]:** domain assumption number xx
- **App:** application
- **Stats:** statistics

1.4 Revision History

- **9/12/19 - Version 1.0:** First release ;

1.5 References

- Specification document “SafeStreets Mandatory Project Assignment”
- IEEE Standard for Information Technology—Systems Design— Software Design

1.6 Document Structure

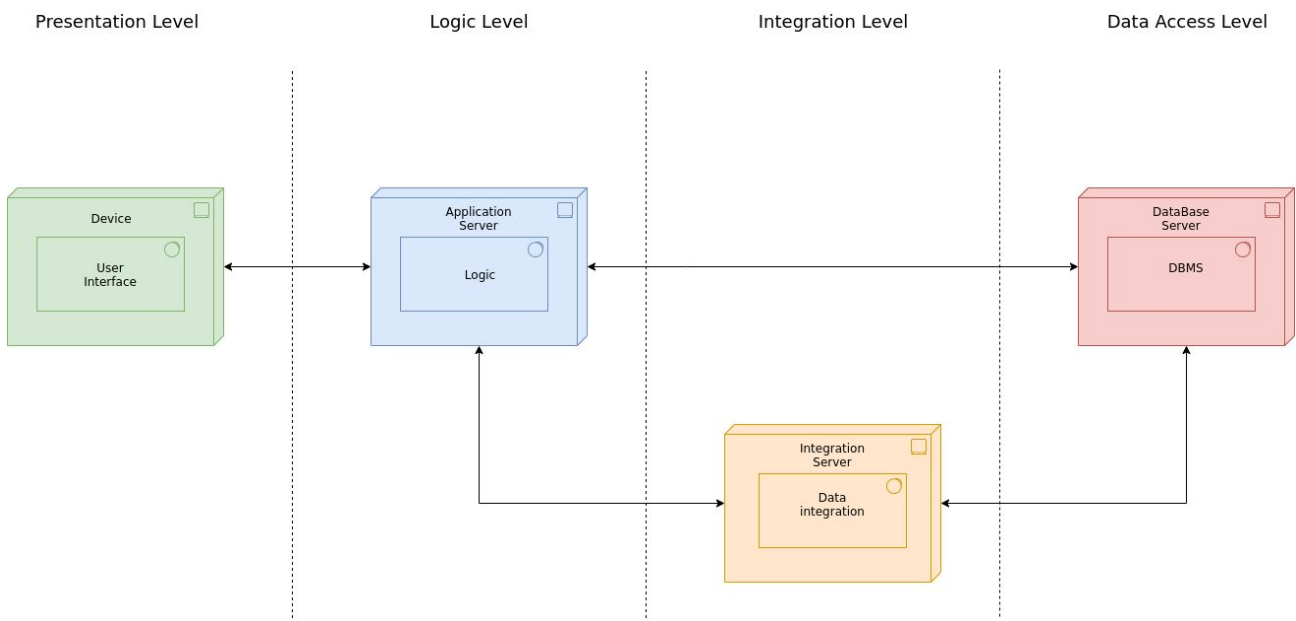
The document is divided in seven parts:

1. **Chapter 1: Introduction.** In this chapter it is given an overall view of the document, its goal and the information needed to understand its contents;
2. **Chapter 2: Architectural Design.** In this chapter it is provided a description of the system's structure, in particular its components, its interfaces, its deployment and the design choices underlying it;
3. **Chapter 3: User Interface Design.** This part will explain how user interface mockups presented in the RASD document were designed;
4. **Chapter 4: Requirement traceability.** This section will explain how the requirements that have been explained in the RASD are satisfied and compliant with the components designed and presented in this document;
5. **Chapter 5: Implementation, integration and tests plan.** In this chapter it is explained the strategy for the implementation, deployment, integration and testing of the various system components.
6. **Chapter 6: Effort Spent.** In this section it is shown the commitment of each group member to the project;
7. **Chapter 7: References.** Here are listed the references taken into account during the project development

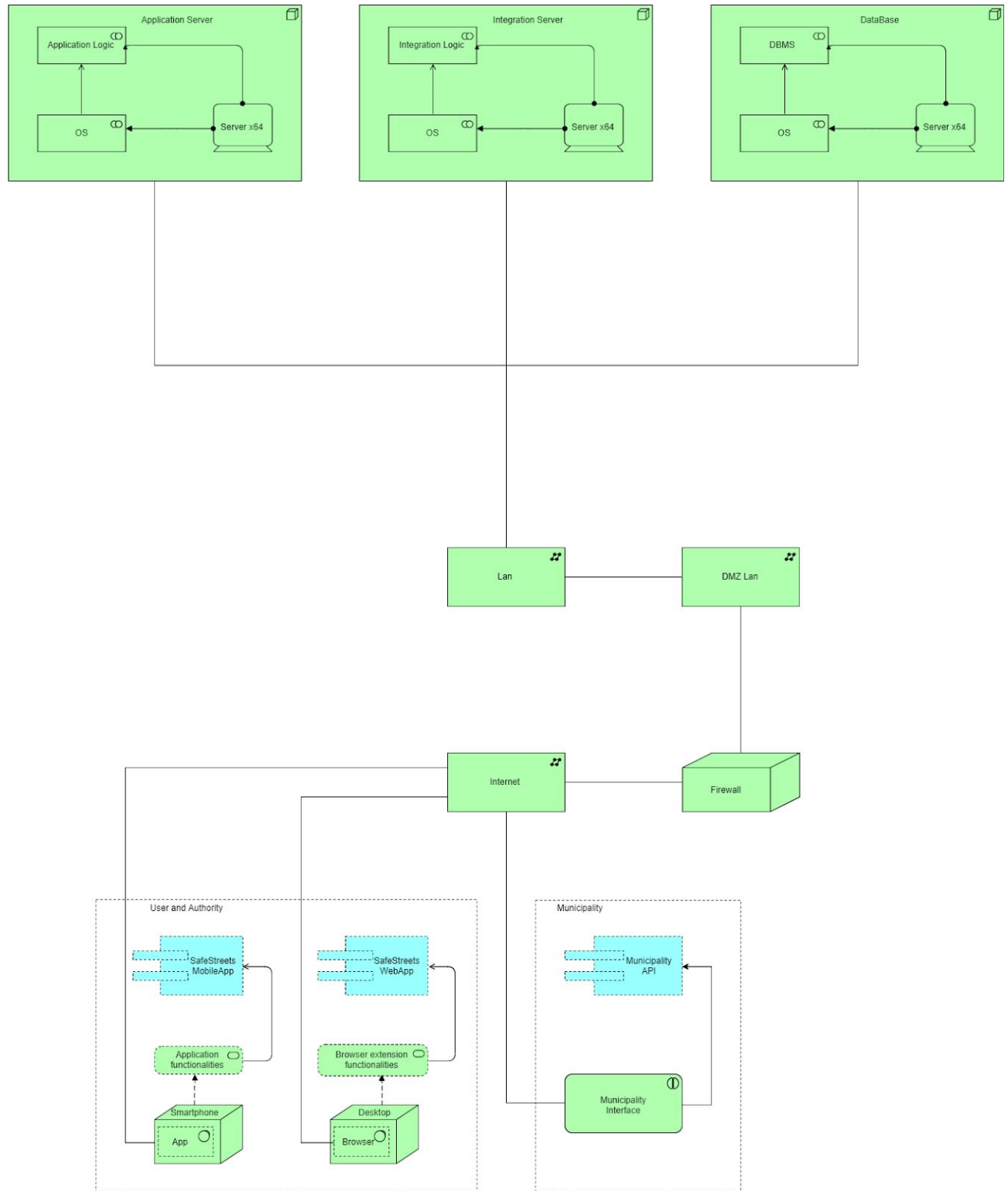
2. Architectural design

2.1 Overview

The application will feature a 4-tier architecture: the presentation level will be responsible for presenting the data to the user and handling his inputs, the logic level will implement all the application core functionalities, the data access level will be aimed at storing all the needed information for the system functioning and finally the integration level will be responsible for merging the data coming from the external sources with the local database.

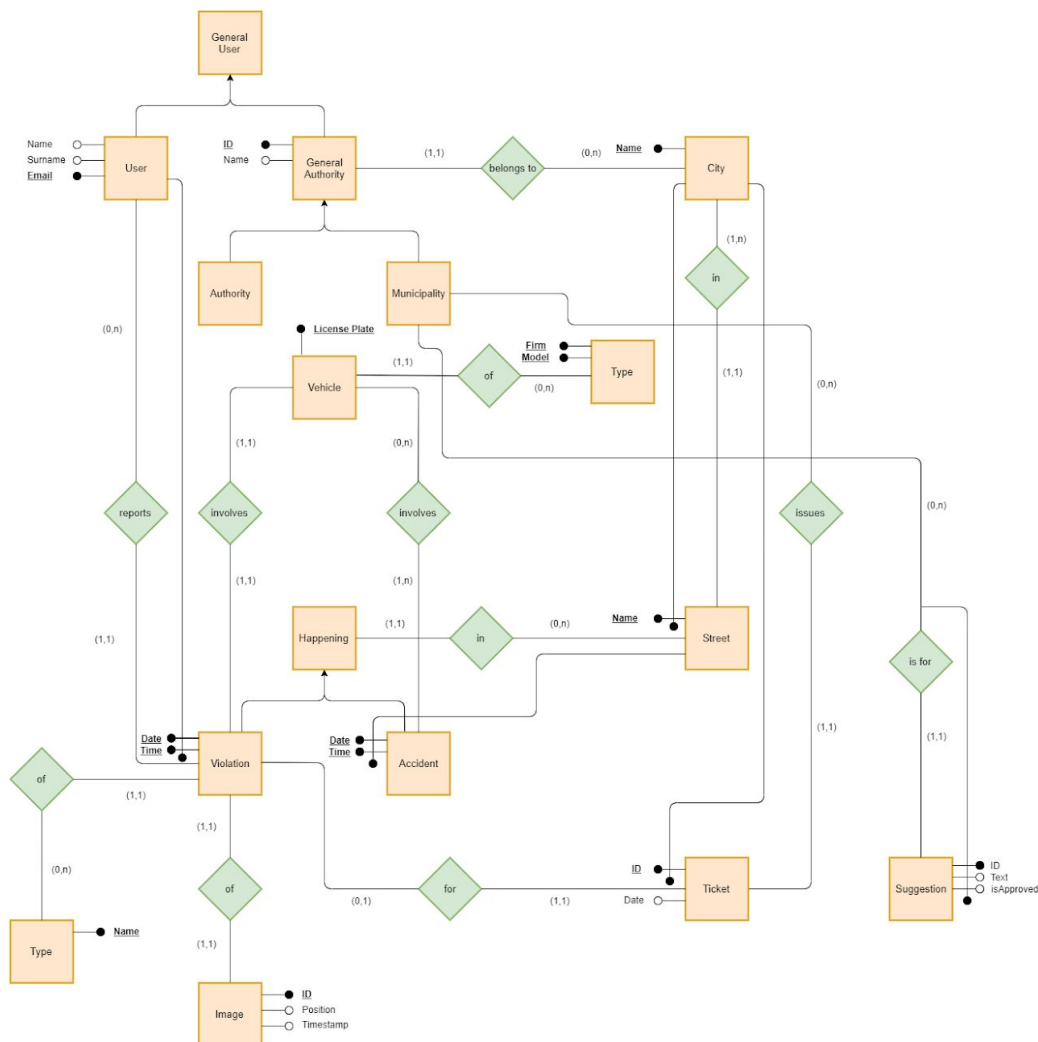


Entering into detail on the architecture, it is provided an ArchiMate representation of the system.



2.2 Component view

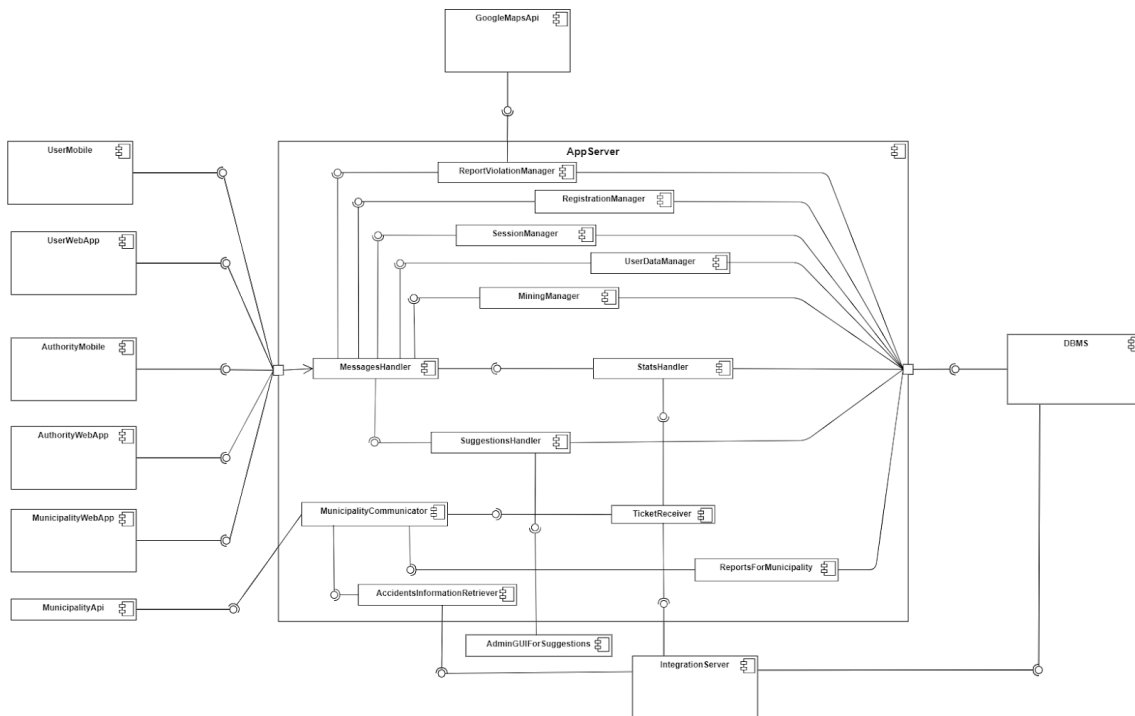
We will describe the components following the approach we used in designing the system, i.e. starting from the model, namely the E-R diagram, implemented through a DB and a DBMS.



Going from the top to the bottom, the E-R diagram features the General User entity which represents a generic application user, be it a citizen or a policeman. General User can be furthermore specialised into User and General Authority. General Authority, in turn, can be specialised in Authority and Municipality. General Authorities are linked to a City which is made up of Streets. Users are the Violation reporters. Each Violation has a predefined Type, a linked Image

and happens in a Street. Also Accidents happen in a Street, for this reason both Violation and Accident have been generalised into the Happening entity. Tickets are issued by Municipalities. Each suggestion is associated with a particular Municipality.

Moving to the Application Server, it is made up of the following components:



- **HandlerMessages:** It receives all messages from clients and depending on the id of them forwards them to the appropriate component to perform a specific functionality. It acts as an event dispatcher. Once the app has finished executing the service the handler returns to the client a reply message containing the outcome of the operation. This component also takes care of keeping the ports open for listening so that clients can connect by opening the connections necessary for communication.
- **ReportViolationManager:** It manages the operations concerning the creation of reports by users analyzing the photos, obtaining the license plate of the vehicle and then storing all the information in the Database interacting with the DBMS.

- **RegistrationManager:** Manages all the registration of an account both for user, authority and municipality, and also it makes the necessary checks.
- **SessionManager:** Manages login and logout operations for both user, authority and municipality and manages all connected account sessions.
- **UserDataManager:** It responds to users' requests to see their information stored by the app by extracting them from the database and returning them to the requesters. This information includes both account information and all reports made.
- **MiningManager:** It manages all the mining requests from both the user and the authority. It extracts from the database all the information of the reports made in a specific area and depending on whether the applicant is user or authority selects the information to be shown, integrates them and then sends them back to the applicant.
- **StatsManager:** Component that provides the interface necessary to modify the statistics of SafeStreet. He goes to modify the data on the statistics present in the database.
- **MunicipalityCommunicator:** Component that opens connections with municipalities in order to interact with their APIs. The component knows all the IP addresses of the registered municipalities that use the advanced SafeStreet services concerning the sharing of information about accidents and the creation of tickets.

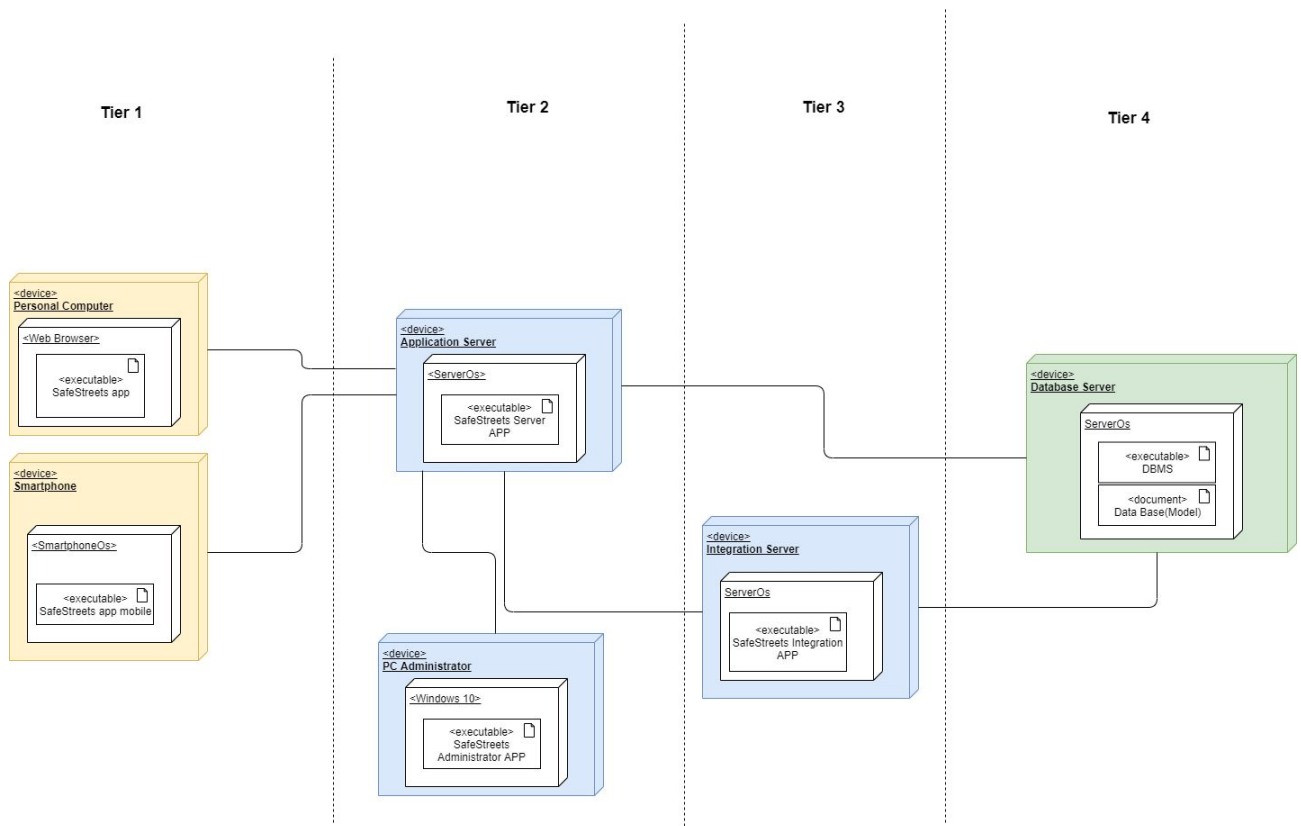
- **AccidentsInformationRetriever:Component** that manages the operations concerning the reception of data on accidents coming from municipalities. The component manages timers for the various municipalities, as soon as it expires it interacts with the **MunicipalityCommunicator** to communicate with a Municipality and then once the data is received in a format (which can be heterogeneous for different municipalities), forward them to the **IntegrationServer**.After the incidents have been integrated, it warns the **Suggestions Handler** to have updated the DB so it can start creating suggestions.
- **AdminGUIForSuggestions:** Is a component that communicates with the **SuggestionsHandler** and allows an administrator to see the suggestions(through a GUI) automatically created by the **SuggestionsHandler** stored in the DB and then the administrator can correct or refine them and then approve them by sending them back to the **SuggestionsHandler**.
- **SuggestionsHandler:**It manages the automated creation of suggestions to be sent to the municipalities and the reading and writing operations for them. The component responds to requests to read from the admin and to modify suggestions for a municipality in the database.It also communicates with the **MunicipalityCommunicator** to send to a municipality all the suggestions approved by the administrator through the **AdminGUIForSuggestions**.
- **ReceiverTicket:** Component that manages the operations concerning the reception of data on tickets coming from municipalities.The component manages timers for the various municipalities, as soon as it expires it interacts with the **MunicipalityCommunicator**. It takes the data of the

tickets coming from the MunicipalityCommunicator that arrive in a specific format and forwards them to the integration server.

- ReportsForMunicipality: Manages the timers associated with the municipalities for periodic sending of updated user reports. When a timer expires, it takes the information on violations(reports) of a specific city and forwards them to the MunicipalityCommunicator who will send them to the municipality of competence.
- IntegrationServer: It mainly deals with receiving tickets and accident data in a specific format, recognizing it and then translating it into a single format so as to avoid conflicts by taking only the important information for the Database. In this way the integration is done correctly.

2.3 Deployment view

Below it is represented the SafeStreets deployment diagram: it shows the execution architecture of the system and represents the distribution (deployment) of software artifacts to deployment targets (nodes). Artifacts, in general, represent pieces of information that are used or are produced by a software development process and are deployed on nodes that can represent either hardware devices or software execution environments.



In the diagram '`<<executable>>`' and '`<<document>>`' are standard UML stereotypes that apply to artifacts: the former represents a program file that can be executed on a computer system while the latter represents a generic file that is not a `«source»` (another UML standard stereotype) file or `«executable»`, in this case it represents the tables in the database.

The four tiers contains:

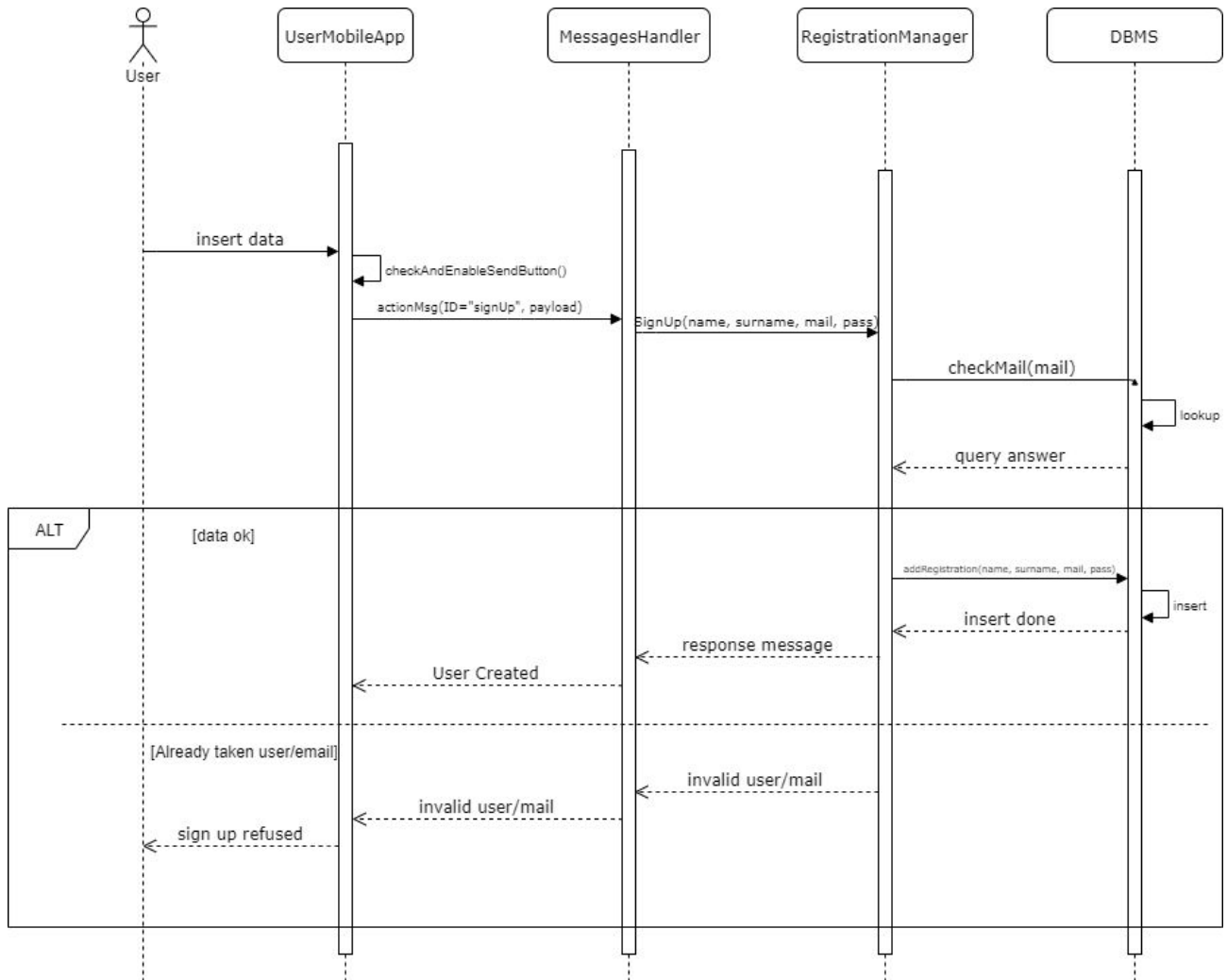
- 1) Tier 1: It contains all the presentation logic and allows both user, authority and municipality to use the SafeStreets services with the aid of a

convenient GUI. In the case of a smartphone the tier is deployed through a mobile app available for both IOS and Android. Instead in the case of a personal computer the tier is deployed through a browser app available for chrome and Mozilla.

- 2) Tier 2 : Contains two devices. In the first (Application Server) the software that contains the logic for all the features of SafeStreets is deployed. In the second one (PC administrator) instead there is a software with GUI that allows administrators to see the suggestions, modify them and then send them to the Server.
- 3) Tier 3: It contains a complex software that receives incident and ticket data in a specific format from the application server and then, by converting them into a single format, integrates them by communicating directly with the DBMS.
- 4) Tier 4: Contains the DBMS that provides an interface to access SafeStreet data contained in the document that constitutes the Data Bases.

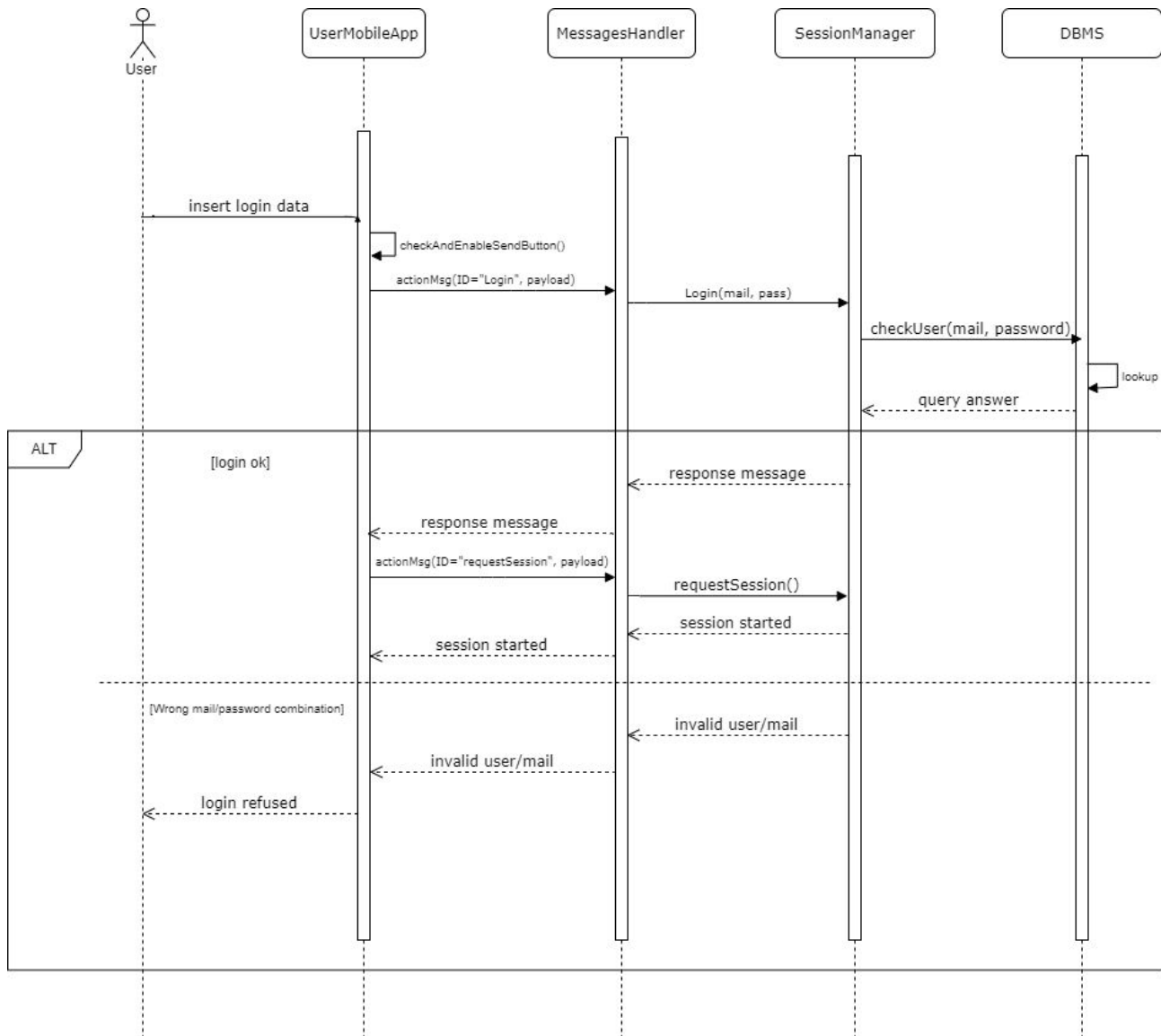
2.4 Runtime view

2.4.1 User SignUp



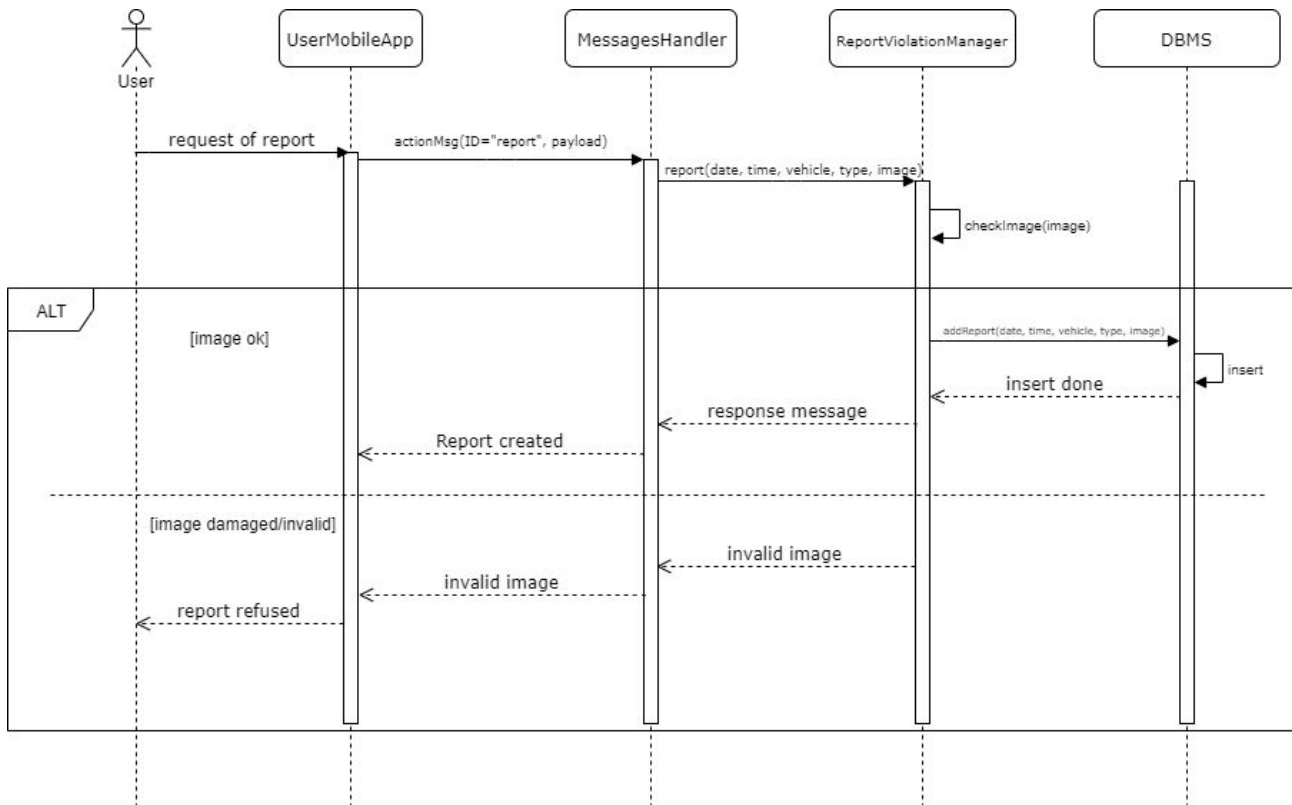
In this diagram it is shown the process through which a user signs up. The user fills the application page with his data. The application checks the validity of the data and then sends them through the MessagesHandler to the RegistrationManager. The RegistrationManager checks the non-existence of the mail in the DataBase. Once verified this, if the answer is positive it proceeds in inserting the user into the DataBase. In the other case it returns an error to the user.

2.4.2 User Login



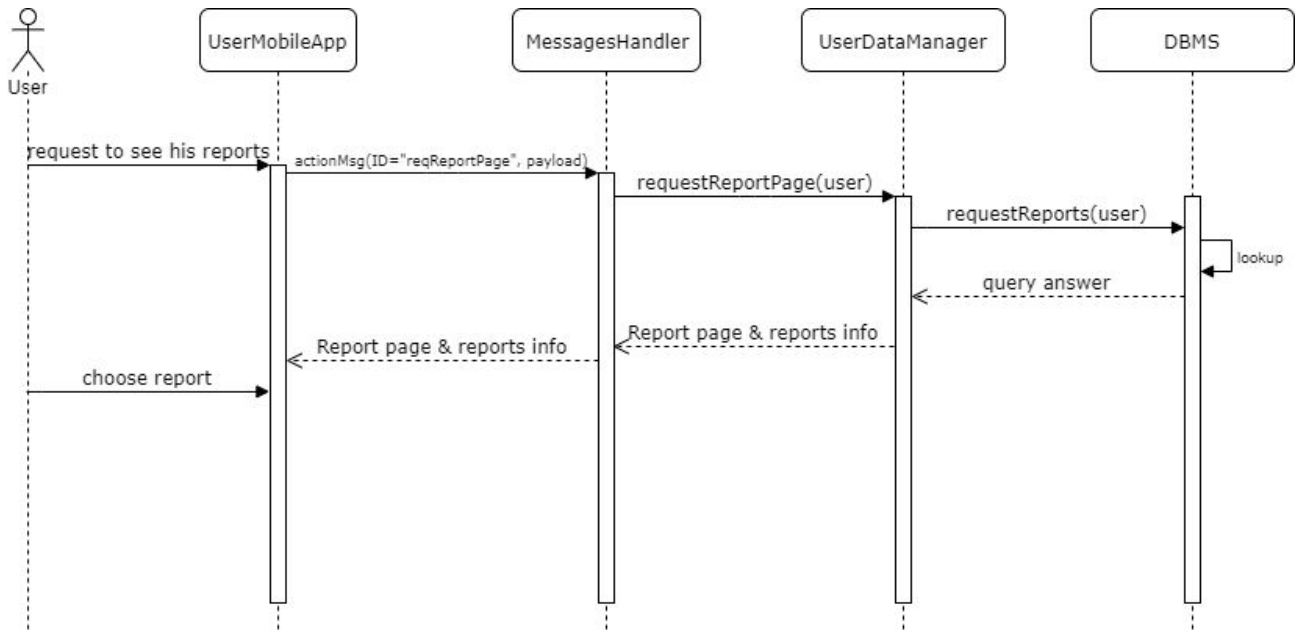
In this diagram it is shown the process through which the user logs in. The user fills the application page with his data. The application checks the validity of the data and then sends them through the MessagesHandler to the SessionManager. The SessionManager checks the correctness of the mail and password combination through the DBMS. Once obtained the answer, if the combination is correct the MobileApp asks the SessionManager through the MessagesHandler to initialize the session. In the case of wrong mail and password combination, the SessionManager returns an error to the user.

2.4.3 User Report



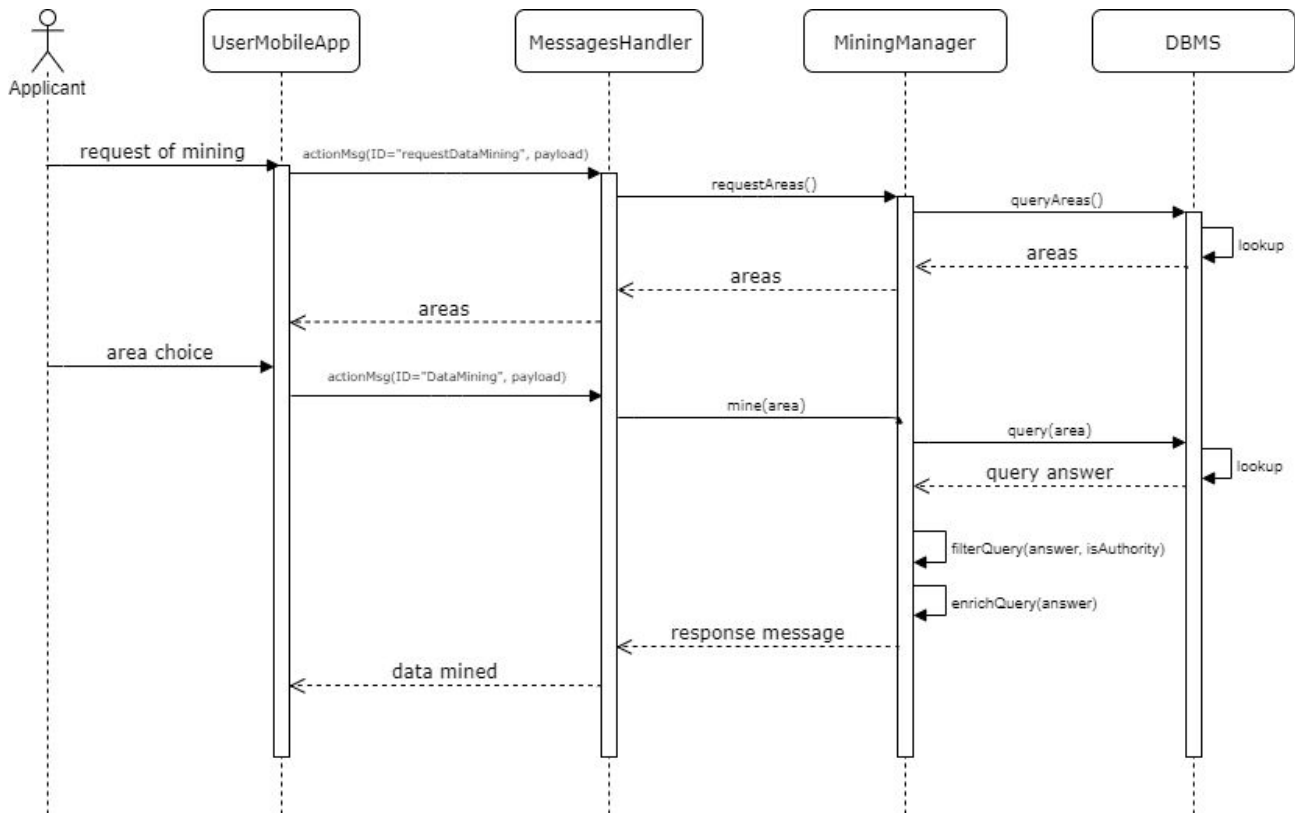
In this diagram it is shown the process through which the user sends a report. The application sends the report to the ReportViolationManager through the MessagesHandler. The ReportViolationManager checks the integrity and validity of the image and if no error occur proceeds in adding the report to the DataBase through the DBMS. In case of invalid or damaged image, instead, it returns an error message to the user.

2.4.4 User See Personal Data



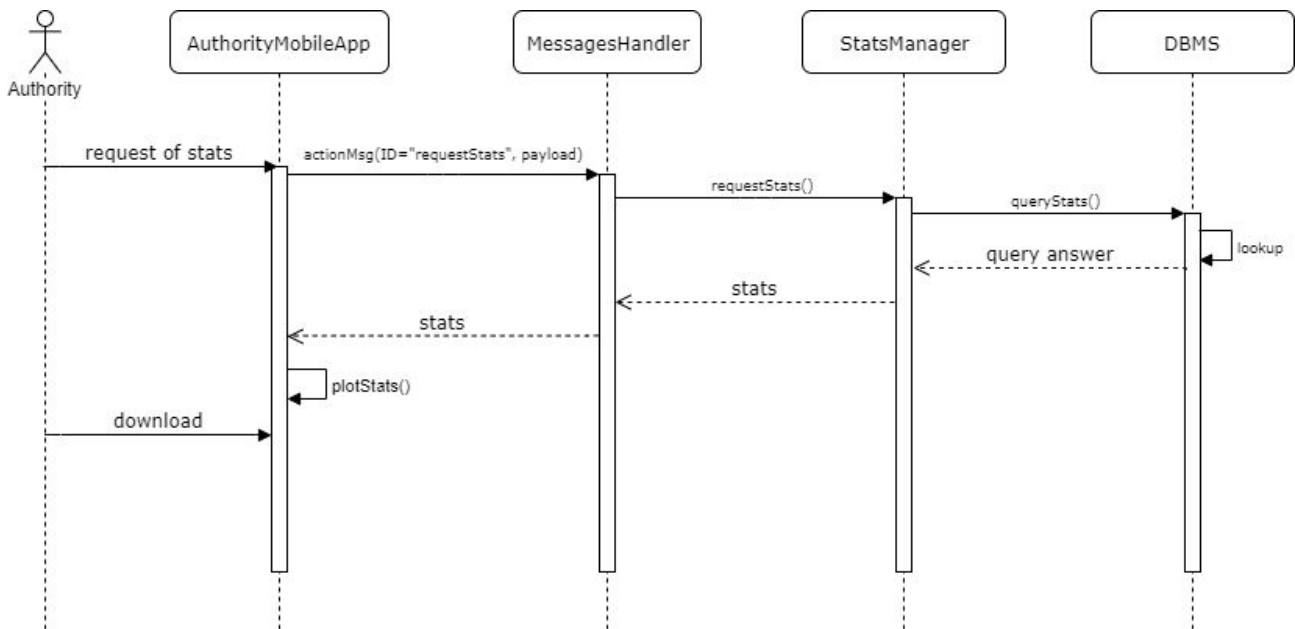
In this diagram it is shown the process through which the user sees his personal reports. The user opens up the corresponding menu page, the MobileApp asks the UserDataManager a list of the user's reports through the MessagesHandler and displays them once received. In case of no reports, nothing is shown.

2.4.5 User Data Mining



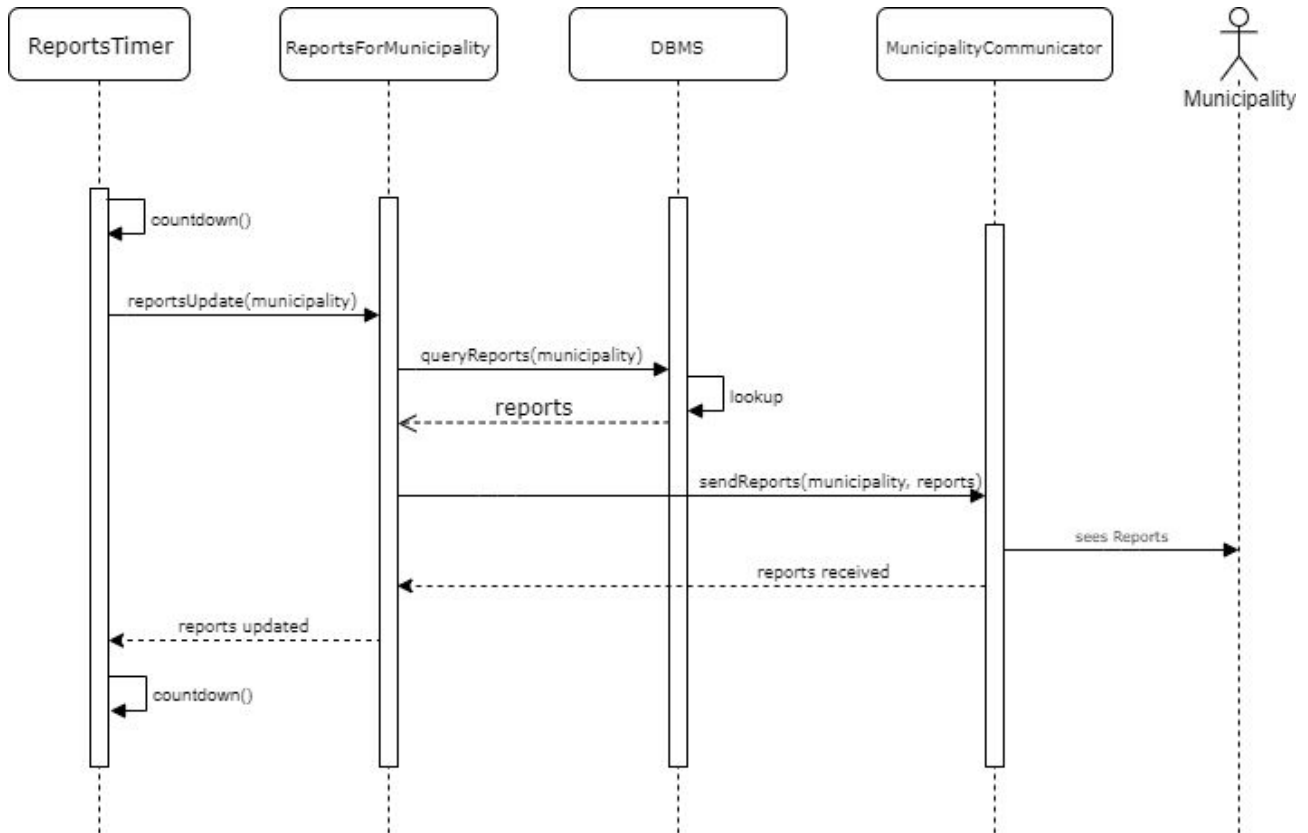
In this diagram it is shown the process through which the user mines data from the SafeStreets DataBase. The MobileApp retrieves a list of the minable areas contacting the MiningManager through the MessagesHandler. The Mining Manager queries the DataBase and returns the list of areas. The MobileApp displays them, possibly sorting them by City. At this point the user chooses an area of interest. The MobileApp contacts again the MiningManager which filters the information queried from the DataBase according to the privileges granted to the user and enriches the answer, for example by selecting some highlighting colours for each area. The answer is then sent back to MobileApp which displays it.

2.4.6 Authority See Stats



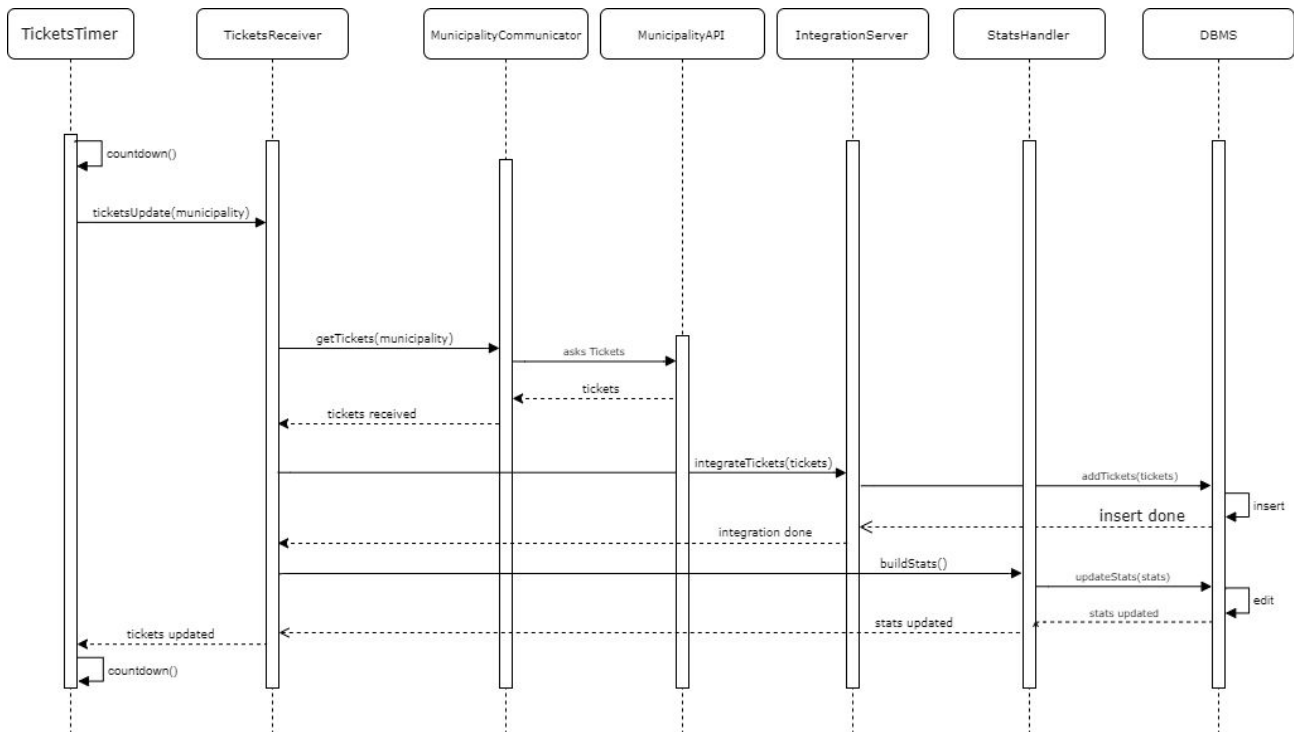
In this diagram it is shown the process through which the Authority sees the SafeStreets stats. The MobileApp contacts the StatsManager through the MessagesHandler. The StatsManager asks for the stats through the DBMS and then returns them to the MobileApp. The MobileApp then proceeds in plotting and displaying them.

2.4.7 Municipality Receives Reports



In this diagram it is shown the process through which the Municipality receives the SafeStreets reports. When a timer linked to the urge to update the reports for a specific Municipality expires, ReportsForMunicipality queries the DataBase and collects the newly created reports. Then it sends the reports through the MunicipalityCommunicator to the Municipality. At this point the timer linked to that particular Municipality starts over again.

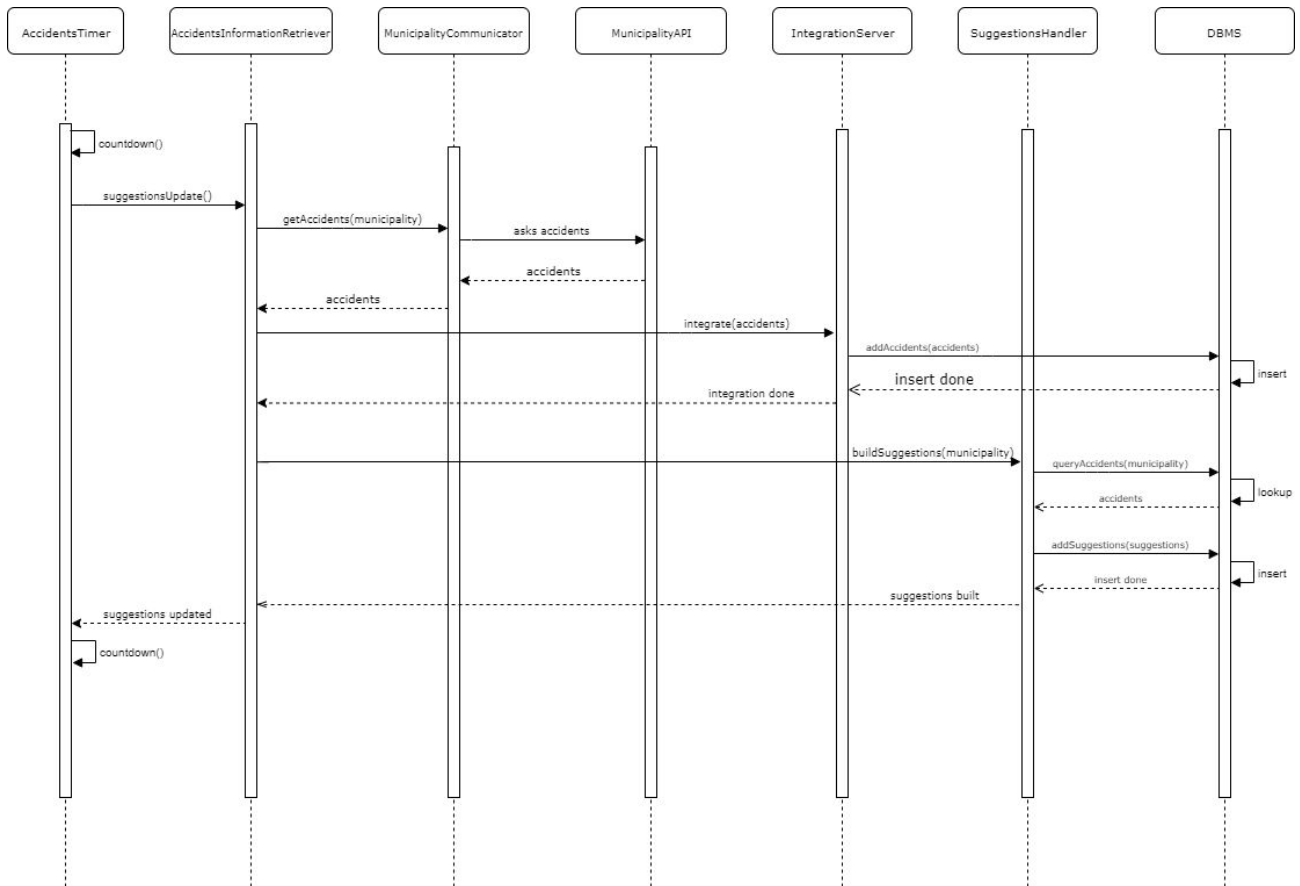
2.4.8 Municipality Sends Tickets



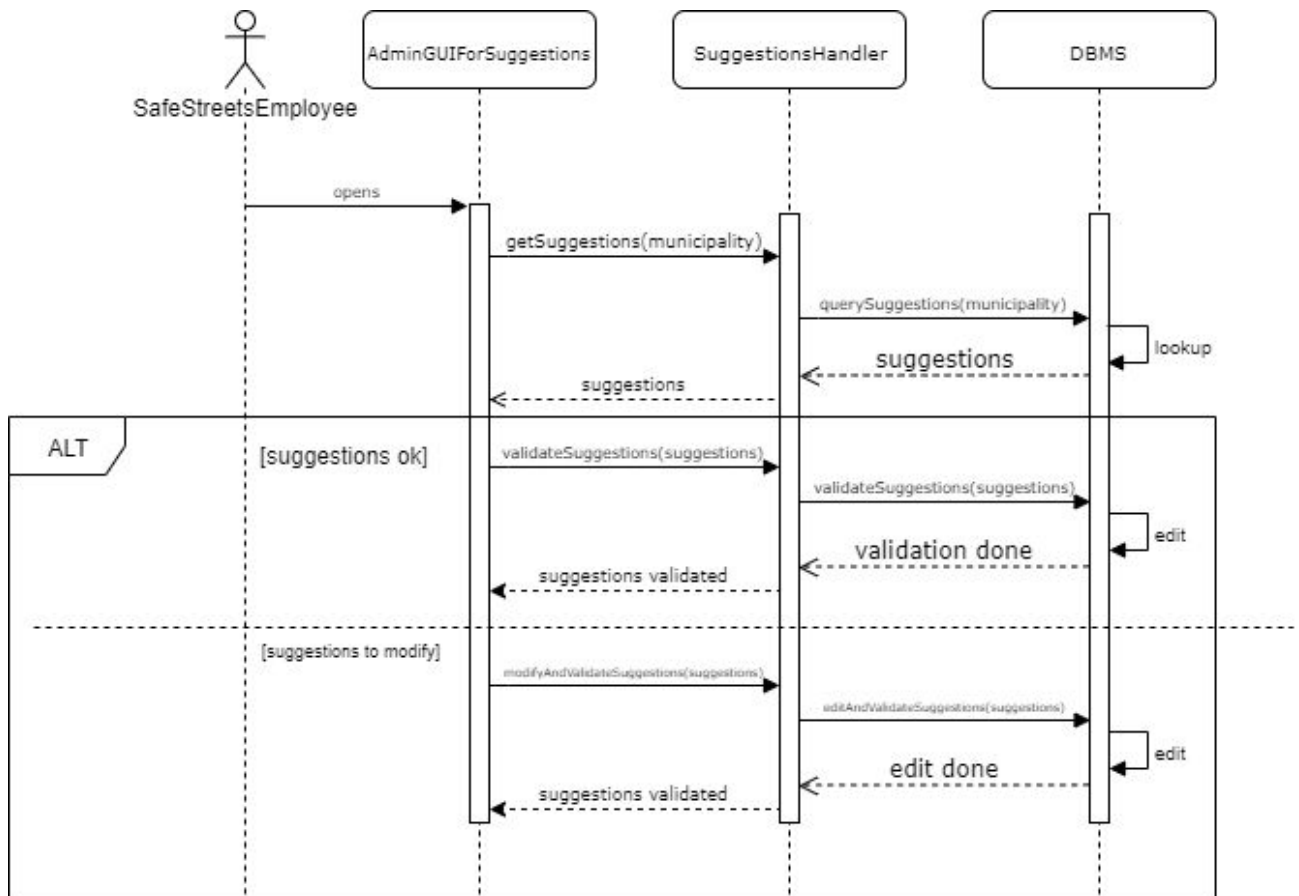
In this diagram it is shown the process through which the Municipality sends the violation tickets to SafeStreets. When a timer linked to the urge to update the violations issued from a specific Municipality expires, TicketsReceiver collects the newly issued tickets from the MunicipalityAPI through the MunicipalityCommunicator. Then it sends them to the IntegrationServer which provides for adding them to the DataBase. Once the integration is done, the TicketsReceiver is notified and the timer linked to the aforementioned Municipality starts over again.

2.4.9 Municipality Receives Suggestions

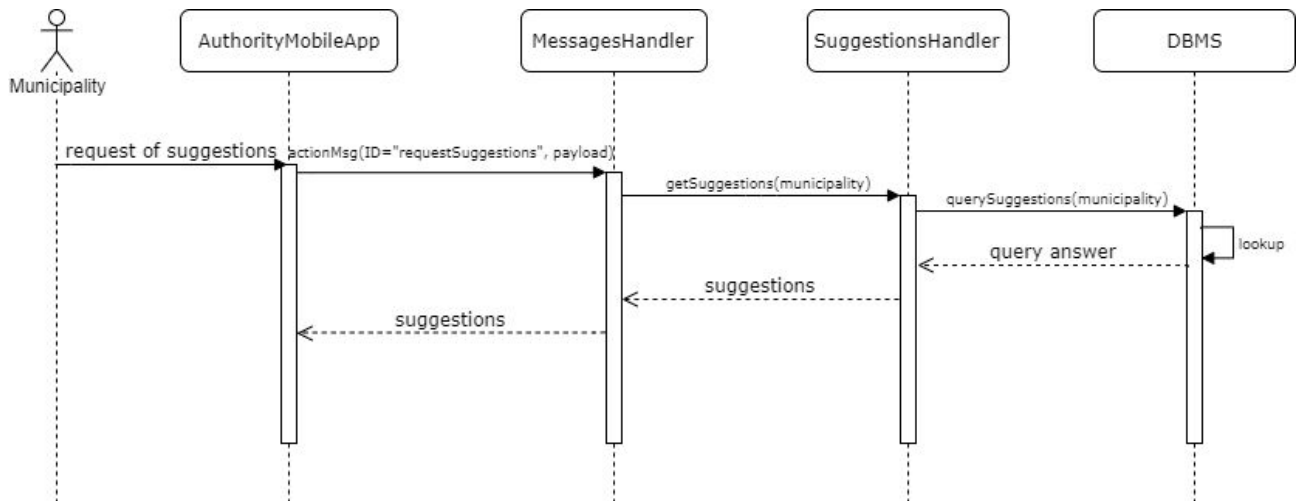
This use case has to be split in three sequence diagrams.



In this diagram it is shown the process through which SafeStreets builds fresh suggestions for a Municipality. When a timer linked to the urge to update the accidents information expires, the AccidentsInformationRetriever component retrieves the new accidents data from the MunicipalityAPI through the MunicipalityCommunicator. Once collected the data, it sends them to the IntegrationServer which proceeds in integrating them into the DataBase. At this point the AccidentsInformationRetriever urges the SuggestionHandler to retrieve the newly added information and use them to build new suggestions which are immediately added into the DataBase. Once this process ends, the timer starts over again.



In this diagram it is shown the process through which SafeStreets employees approve the suggestions automatically created by the system. The AdminGUIForSuggestions proceeds in retrieving the not-yet-approved suggestions from the DataBase through the SuggestionsHandler and depending on their quality there are two options. Either they are all correct and thus the components validates them or some of them need to be reviewed and thus edited and then validated.



In this diagram it is shown the process through which the Municipality accesses the suggestions SafeStreet crafted for it. The MobileApp contacts the SuggestionsHandler through the MessagesHandler and retrieves the suggestions. The MobileApp then proceeds in displaying them.

2.5 Component interfaces

In the above figure the component interfaces belonging to the application server, are represented. The dependency relation are omitted because they are already shown in the Component Diagram. Furthermore, the logical translation of the E-R diagram is also shown. The classes on the left represents the tables of the DB with their attributes. The use relationships¹ between the DB data and the interfaces of the application server components are also shown.

Possibili osservazioni

1. Mancanza dei tipi di ritorno

¹ In case of plural parameters, those are meant to be lists

2.6 Selected architectural styles and patterns

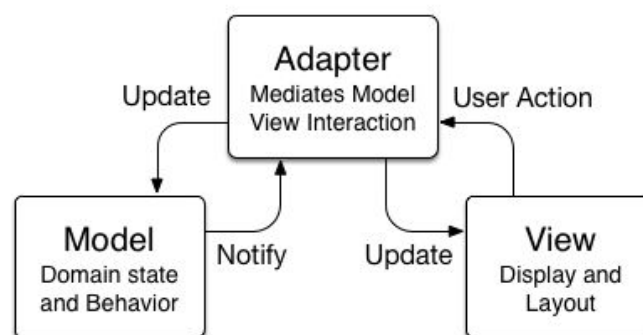
4 Tier Client-Server

The application will have a multilayered architecture. We distinguish between:

- Presentation Tier: this is the layer responsible for presenting the content retrieved from the application server. It involves either citizens and authorities devices;
- Application Tier: this is the layer responsible for the logic of the system, here information is processed, elaborated and combined;
- Integration Tier: this is the layer needed in order to integrate the data coming from external sources (Municipalities' API and so on) with the internal stored information;
- Data Access Tier: this is the layer in which information is stored.

This solution, which is an extension of the classical 3 tier Client-Server Architecture, allows a better flexibility and an easier implementation, deployment and testing.

Model View Adapter



Given the choice of a relational DB we opted for the Model View Adapter architectural pattern in order to avoid a direct contact between View and Model. The Model is represented by the DataBase, the View by the

application devices and the Adapter by the Application and Integration Servers.

Event-Based System

The core component of the Application Server is the MessagesHandler which will be implemented as a centralized event dispatcher. It will collect the subscriptions coming from the various “Manager” components linked to it and forward the proper messages to them.

Thin Client

Given the need for image validation, which can’t obviously be performed locally, the system will feature a thin client. It will be easier to deploy, lighter and will face less security threats.

Relational DB

The Model will be implemented with relational DB. This choice will ensure a great coherence and low redundancy. Performances will not be an issue given the fact that this particular scenario does not require them.

Loose Coupling

The system will be loosely coupled: this choice will minimize the unnecessary interactions between components and ease the maintenance process. The system will be less performing in data synchronization but the application isn’t intended to be real time, so the choice is appropriate.

Stateless components

Given the presence of a relational DB, each component does not need to store an internal state. This justifies the choice of a stateless component pattern.

2.7 Other design decisions

Google APIs

Google is the designated provider for the geolocalisation functionalities. In particular the application will employ the Google Maps API to provide the information needed for the violation report.

Municipality API

The application will be interfaced with Municipalities through their APIs. Given the fact that each API will presumably be different, this is a key aspect which must be managed by a dedicated component, namely the Integration Server.

AdminGUIForSuggestions

This GUI is needed to validate the automatic generated suggestions. Given the importance that some of those suggestions could have and the strategic role they have in the SafeStreets environment, it is a judgement call.

3. User Interface Design

Since the application mocks have already been inserted in the RASD, the UX diagrams will be integrated in the 2.0 revision of the document above and purposely omitted here for better cohesion.

4. Requirements traceability

4.1 Functional Requirements

In order to demonstrate that the built system is able to fulfill the functional requirements defined in the RASD here is presented a mapping between those requirements and the system components²:

- **R01:** Safestreets take the license plate from the photo sent by user.
 - ReportViolationManager
- **R02:** The system must allow the user to insert the type of violation.
 - ReportViolationManager
- **R03:** Safestreet stores the reports provided by users if the photo is not broken.
 - ReportViolationManager
- **R04:** The system must use a gps system on the device that makes the signal to obtain the position of the violation.
 - ReportViolationManager
- **R05:** The system must save the collected data of users registered to SafeStreets in real time.
 - RegistrationManager
 - ReportViolationManager

² Session Manager and Message Handler are needed for all the functional requirements and thus not included.

- **R06:** The system must allow the user and the authority to access his own registered data and all the reports done with the possibility to consult them.
 - UserDataManager
- **R07:** The system must allow the applicant to select the interested areas where the parking violations occurred.
 - MiningManager
- **R08:** The system must show the extracted data to the applicant.
 - MiningManager
- **R09:** If the applicant is a user the system must show only the areas with the highest parking violations frequency.
 - MiningManager
- **R10:** If the applicant is an authority the system must show the areas with the highest parking violations frequency but also what kind of vehicles committed violations often and the license plates of the most recent infractions.
 - MiningManager
- **R11:** The system must extract the data about the accidents of a territory from the municipality.
 - AccidentsInformationRetriever
- **R12:** The system must extract the data about the accidents periodically.
 - AccidentsInformationRetriever
- **R13:** The system must send the suggestions to the municipality.
 - SuggestionsHandler

- **R14:** The system must create suggestions for the municipality depending on the information received.
 - SuggestionsHandler
 - AdminGUIForSuggestions
- **R15:** The system must send the suggestions periodically.
 - SuggestionsHandler
- **R16:** The system must create and make available the information about the violation of a territory for the municipality.
 - ReportsForMunicipality
- **R17:** The system must extract the data about the issued tickets periodically from municipality.
 - ReceiverTicket
- **R18:** The system will use the data about the issued tickets to generate stats about the efficiency of the service.
 - ReceiverTicket
- **R19:** The system must built stats to show the authority the trends in the issuing of tickets,the most egregious offenders and some parameters about the efficiency of SafeStreets to report the violations.
 - StatsManager

4.2 Non-Functional Requirements

In order to demonstrate that the built system is able to fulfill the non-functional requirements defined in the RASD here is presented a mapping between those requirements and the system components:

- **Security:** Security is guaranteed by the SSL protocol which will have ports available precisely to grant secure and encrypted communications. Moreover, the system infrastructure also includes firewalls that will filter the traffic coming from the outside world to prevent unwanted packets from reaching the servers.
- **Maintainability:** The loose coupling pattern applied to all components of the AppServer allows easy maintenance as the components are very independent of each other. Furthermore, the Handler Messages and Integration Server components guarantee flexibility. The first, as it follows the pattern Event Handler, if by chance new functions are added for the clients, it is enough to create a new event and associate it with the call of a function in a specific component. the inclusion of various translators for municipalities who send data with a logical scheme or different format.
- **Portability:** The components UserMobileApp, UserWebApp, AuthorityMobileApp, AuthorityWebApp, MunicipalityWebApp and MunicipalityMobileApp guarantee portability. You can use SafeStreet services both via the browser web app and the mobile application.

5. Implementation, integration and test plan

Identify here the order in which you plan to implement the subcomponents of your system and the order in which you plan to integrate such subcomponents and test the integration.

NOTA: rimarcare il fatto che avendo una struttura a quattro tier il deploy è semplificato (decoupling di app e integration server facilita il testing)

Elencare subcomponents

Per ogni subcomponent elencare i manager

Ipotizzare roadmap implementazione

Aggiungere metriche (code duplication, coverage, ecc)

Vedere 3 - DD per jenkins e pattern testing

6. Effort Spent

Matteo Ferrini

Description of the Task	Hours
Introduction	99
Architectural Design	99
(User Interface Design)	99
Requirements Traceability	99
Implementation, Integration and Test Plan	99

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7. References